

Input & Output Decorators in Angular



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Input Decorator (@Input)

- The @Input decorator allows a parent component to pass data into a child component.
- It makes a property of a child component available for binding by a parent component
- If we have relationship between component and we want to send the data from parent component to child then we can use @Input() decorator

child.component.ts

```
import { Input } from '@angular/core';

@Component({
  selector: 'app-child',
  template: `<div>{{ childProperty }}</div>`
})
export class ChildComponent {
  @Input() childProperty: string;
}
```

How to Use Input Decorator

1. Define the property in the child component with @Input decorator.
2. Bind the property in the parent component using property binding ([]).

```
<!-- Parent Component Template Selector Reference Here -->
<app-child [childProperty]="parentData"></app-child>
```

Output Decorator (@Output)

- The @Output decorator allows a child component to emit events to a parent component.
- It creates custom events that a parent component can listen for.
- If we want to send the data from the child component to the parent then we can go with @Output() with the help of EventEmitter .

child.component.ts

```
import { Output, EventEmitter } from '@angular/core';

@Component({
  selector: 'app-child',
  template: `<button (click)="emitCustomEvent()">Click Me!</button>`
})
export class ChildComponent {
  @Output() customEvent = new EventEmitter<string>();

  emitCustomEvent() {
    this.customEvent.emit('Data from Child');
  }
}
```

How to Use Output Decorator

1. Define the event property in the child component with @Output decorator.
2. Emit the event in the child component when needed.
3. Listen for the event in the parent component using event binding (()).

```
<!-- Parent Component Template -->
<!-- Child component selector reference -->
<app-child (customEvent)="handleCustomEvent($event)"></app-child>
```

Recap

@Input() allows passing data from parent to child.

@Output() allows emitting custom events from child to parent.

They are essential for communication between components in Angular.

Note : If we want to send the data from parent to child we can send only between sibling(parent child relationship) component